

offers opportunities beyond traditional OEMs, with subsystems and components ripe for indigenous design and manufacturing. Electric vehicles represent a promising frontier as they are the most electronic systems. Other than batteries, all other subsystems of electric vehicles are prime for Indian-designed products and manufacturing.

Drones, too, hold immense potential, especially with recent import restrictions. While the assembled drones may be imported, the electronics components like radio controllers, flight controllers, and sensors can be designed and manufactured locally.

Mapping India's capability landscape

For India to succeed as an electronics and semiconductor product nation, a strong engineering and balanced workforce is a must. To ensure we have the right capabilities in all aspects of the electronics and semiconductor value chain, we must assess our current capabilities across the electronics and semiconductor ecosystem to gauge where India stands today.

As depicted in Fig. 4, we have a very good and growing market for electronics and semiconductor products and an excellent infrastructure for electronics manufacturing. Over the last three decades, strong capabilities have been developed for electronics systems design, semiconductor chip design, and embedded software design as a service to the global product companies.

For example, India has over 125,000 chip design engineers working for global semiconductor companies, and it is said that there is hardly any chip which does not have contributions from an Indian chip design engineer. Looking forward to domestic and global requirements, it is expected that India will

Suggested initiatives to make India an electronics and semiconductor product nation

- 1000 electronics products and 100 semiconductor chip domestic companies in next 5 years
- PDLI (Product Design Linked Incentives) scheme for electronics product design with \$500 million outlay, Similar to DLI scheme which was created for semiconductor chips
- PLI incentives should be based on percentage of value addition. Higher value-added manufacturing should get a higher % of PLI incentives. A revision should be considered in the formula to calculate value addition
- DLI and PDLI schemes should have seed funding mechanisms in addition to reimbursement mechanisms in the current DLI scheme.
- Create strong infrastructure for product design, moulding, tooling, and enclosures, which is essential for modern electronics products.
- Create a fund of \$100 million/year for 5 years to support chip tape-outs by students

need to produce at least 125,000 more chip design engineers in the next five years. Similarly, the talent must be developed for electronics products and system design.

With the right focus, India is poised to become a talented supplier to the world for the design and manufacturing of electronics products and semiconductor chips. The government of India has started multiple programmes to accelerate the pace of talent development in India.

Fostering a strong ecosystem for hardware startups in India

Fostering a dynamic ecosystem for hardware startups in the electronics and semiconductor sectors is pivotal for India's ambition to lead in this key industry. Despite substantial

achievements across various segments, electronics and semiconductor startups face significant challenges, contrasting sharply with the success of software startups over the past decade. These software enterprises have notably contributed to India's digital public infrastructure, generating numerous unicorns and leading public firms. However, electronics and semiconductor ventures lag due to higher expenses, extended development times, scarce venture capital, and a lack of successful precedents.

Addressing these hurdles, the government of India introduced the Design Linked Incentive (DLI) scheme under its semiconductor policy. It offered robust support, including fiscal incentives covering 50% of project costs and complimentary access

to premier design tools from leading EDA firms like Synopsys, Cadence, and Siemens. Additionally, states like Odisha augment these benefits with up to 20% extra fiscal incentives, pushing total support to 80-85% of project costs, alongside 4-6% PLI benefits on sales volume. This comprehensive support framework positions the DLI scheme as a globally unparalleled initiative to propel semiconductor startups to new heights of commercial and innovative success.

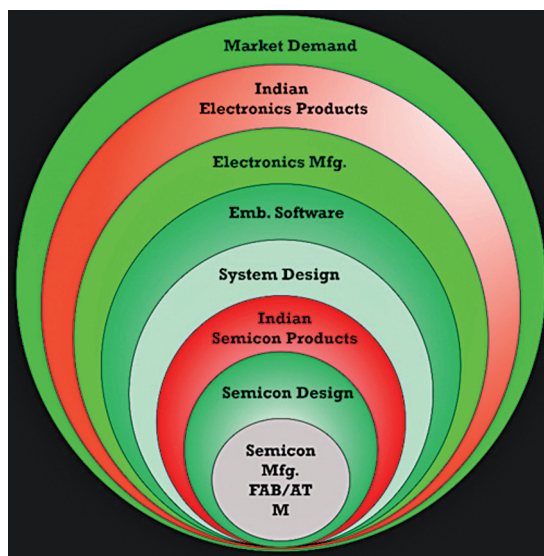


Fig. 4: India's capacity landscape